Roll No. 25325

DRONACHARYA COLLEGE OF ENGINEERING, GURGAON
PRE-UNIVERSITY EXAMINATION MAY- 2024
(GU, GURUGRAM)

Semester: IV(B.Tech.) **Branch: CSE (AI & ML)**

QUESTION PAPER

Subject with Code:R Programming
Time:3 hours **Max Marks: 30**

Note: Question paper contain total Seven questions. Attempt five questions in all, selecting Four question from Section-B. Question No. 1 of Section-A is compulsory. All questions carry equal marks.

SECTION-A

Q. 1. Objective type / Short-answer type questions.

i. Write some advantages of R programming language over Python. Also, discuss the application of R in various fields.
ii. Bar charts vsHistogram vs Scatter plot in R.x
iii. Describe the usage of scan( ) and readline ( ) functions in R?
iv. Explain how arrays are created in R with a suitable example.
v. Explain factors and common functions used with factors.
vi. Explain the getwd() and setwd() functions.
vii. What is the ggplot2 package in R?

SECTION-B

Q. 2. a)Identify the steps in creating and running an R script in different R tools (R studio, R command prompt).
b)Write the rules for naming a variable. Discuss a few important R data structures.

Q. 3. a)Discuss about control statements and loops in R programming with examples.
b)Explain different types of operators in R

Q. 4. a)Explain the following built-in functions in R: - mean(), paste(), sum(), min(), max(), seq(), rep().
b)Describe the following: -
(i) calling a function without an argument
(ii) calling a function with argument values.

Q. 5. a)What are different functions that can be applied on the list, explain with an example.
b)Explain various operations and important functions that can be performed on vectors in R.

Q. 6. a) Discuss in detail about matrices like creating matrices, applying rbind( ) and cbind( ) functions to matrix and matrix operation in R with examples.
b) How do you create and merge a data frame in R? Also, discuss the functions: dim(), nrow(), ncol(), str(), Summary(), names(), head(), tail()apply on the data frame.

Q. 7. a)How to read and analyze the CSV file data using the functions: summary(), min(), max(), range(), mean(), median(), apply().
b)Explain the different types of plots and charts used for data visualization in R?